EXPERIMENT-4

AIM: Create a basic network of one server and two computers and two mobile/movable devices using appropriate network wire and verify the connectivity.

THEORY:-

WIRELESS NETWORK:

A wireless network is a network in which devices communicate and share data without using physical network cables. It uses radio signals, such as Wi-Fi**,** to connect devices like computers, laptops, smartphones, tablets, and servers through a wireless router or access point. Wireless networks provide easy mobility, flexibility, and convenient access to network resources.

TYPES OF WIRELESS NETWORKS:

1) WPAN – Wireless Personal Area Network

WPAN is a wireless network designed for communication between devices within a short distance, usually around a person. It is mainly used to connect personal devices such as smartphones, smartwatches, wireless headphones, keyboards, and mice. Bluetooth and NFC are common examples of WPAN.

2) WLAN – Wireless Local Area Network

WLAN is a wireless network that covers a limited geographical area, such as a home, school, college, office, or computer laboratory. Devices connect through a Wi-Fi router or access point without requiring physical network cables. WLAN is commonly used for sharing files, accessing the internet, and communicating between devices.

3) WMAN – Wireless Metropolitan Area Network

WMAN is a wireless network designed to cover a larger area than a WLAN, such as a town or city. It can connect multiple buildings or local networks within a metropolitan area. WMAN is useful for providing wireless communication across large institutions or public areas.

4) WWAN – Wireless Wide Area Network

WWAN covers a very large geographical area, such as different cities, states, or countries. It uses technologies provided by cellular and telecommunication networks. 2G, 3G, 4G, and 5G mobile networks are examples of WWAN. It allows users to access communication and internet services while moving from one location to another.

5) WSN – Wireless Sensor Network

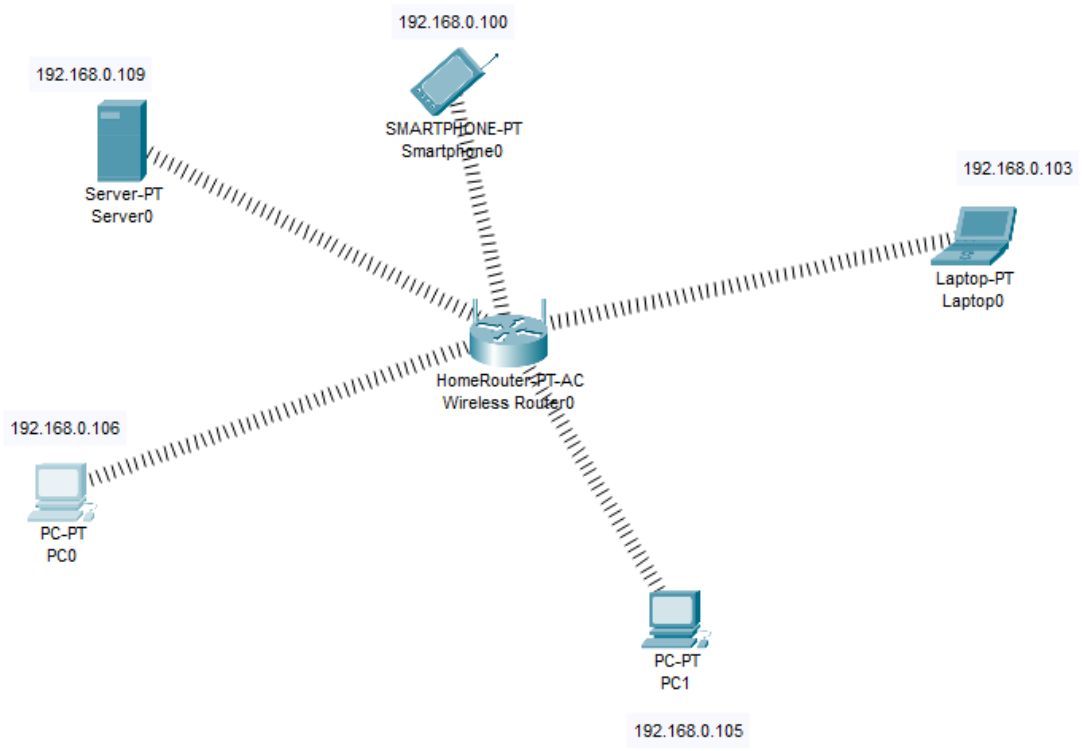
WSN consists of a number of wireless sensors that collect information from their surroundings and send it to a central system. These sensors can measure temperature, humidity, pressure,

motion, and other conditions. WSNs are commonly used in agriculture, healthcare, environmental monitoring, smart homes, and industries.

WAP(WIRELESS ACCESS POINT):

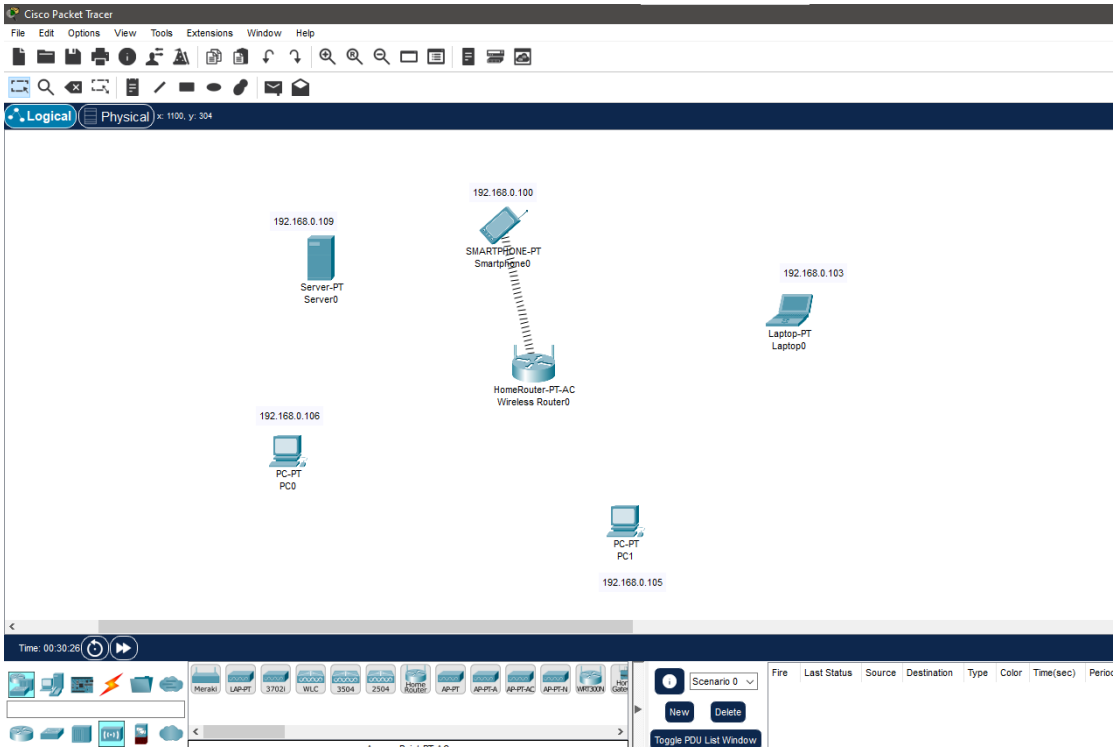
A Wireless Access Point (WAP) is a networking device that allows wireless devices such as laptops, computers, smartphones, and tablets to connect to a wired network using Wi-Fi**.** It receives data from the wired network and transmits it wirelessly to connected devices.A WAP generally works as a central connection point for wireless devices within a particular area. It helps devices communicate with each other and access network resources or the internet without using physical cables.

TOPOLOGY:-

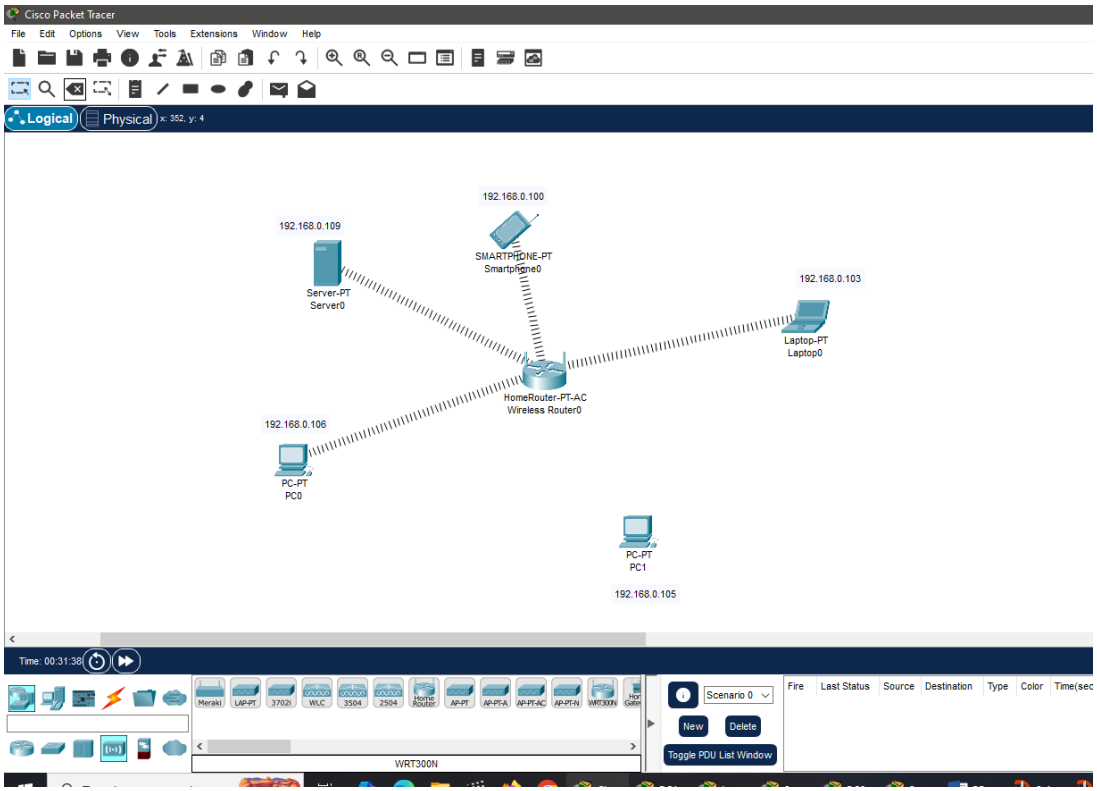


STEPS:-

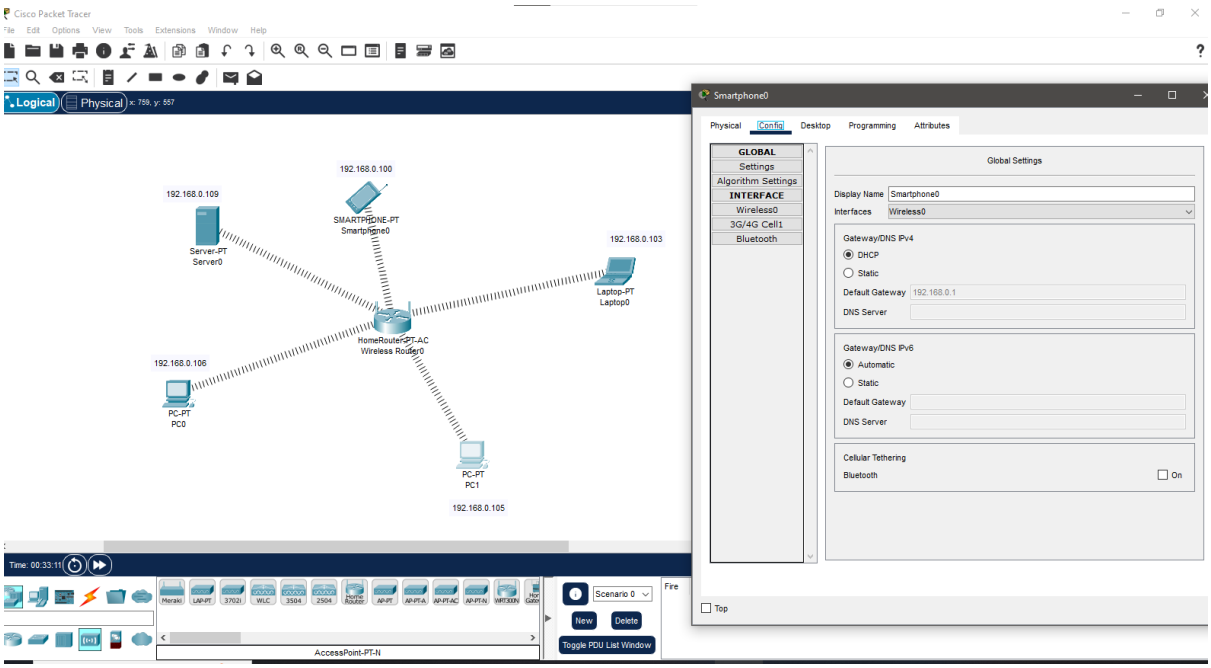
1)


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical
192.168.0.100
192.168.0.109
SMARTPHONE-PT
Smartphone0
Server-PT
Server0
192.168.0.103
Laptop-PT
Laptop0
HomeRouter-PT-AC
Wireless Router0
192.168.0.106
PC-PT
PC0
PC-PT
PC1
192.168.0.105
Time: 00:30:26
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)

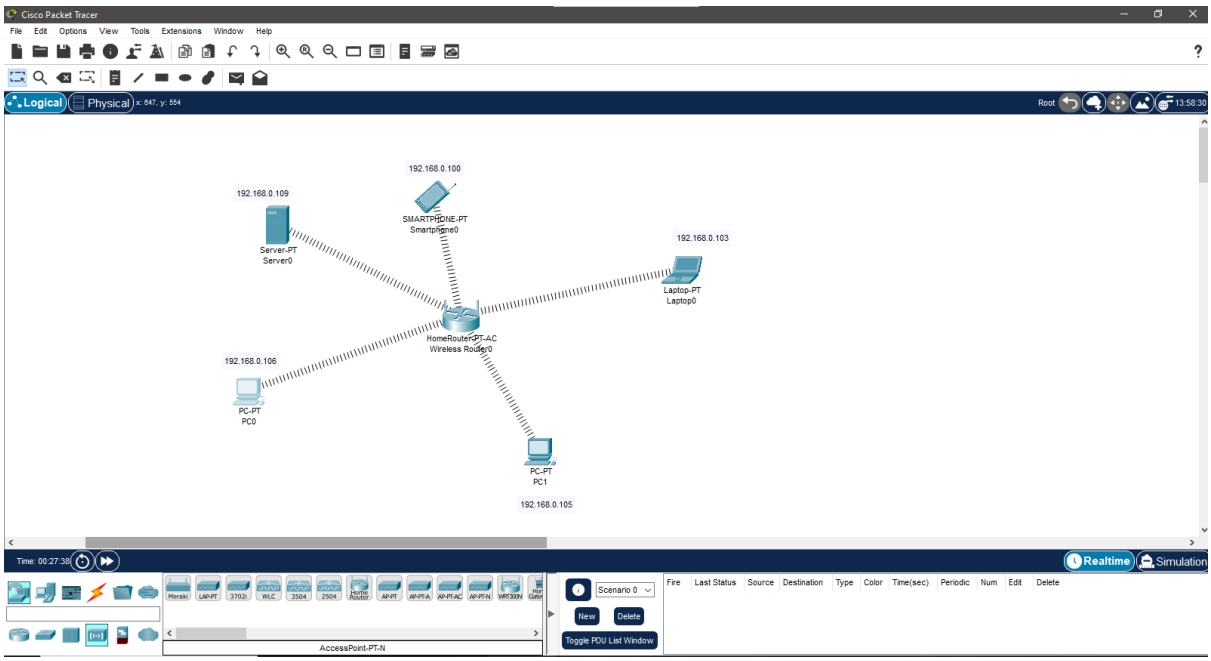
2)


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical
192.168.0.100
192.168.0.109
SMARTPHONE-PT
Smartphone0
Server-PT
Server0
192.168.0.103
Laptop-PT
Laptop0
HomeRouter-PT-AC
Wireless Router0
192.168.0.106
PC-PT
PC0
PC-PT
PC1
192.168.0.105
Time: 00:31:38
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
WRT300N

3)


Cisco Packet Tracer
Logical
Physical
192.168.0.100
SMARTPHONE-PT
Smartphone0
192.168.0.109
Server-PT
Server0
192.168.0.103
Laptop-PT
Laptop0
HomeRouter-PT-AC
Wireless Router0
192.168.0.106
PC-PT
PC0
PC-PT
PC1
192.168.0.105
Smartphone0
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
Wireless0
3G/4G Cell1
Bluetooth
Global Settings
Display Name
Smartphone0
Interfaces
Wireless0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Cellular Tethering
Bluetooth
On
Top
Time: 00:33:11
Scenario 0
New
Delete
Toggle PDU List Window
AccessPoint-PT-N

RESULT:-


Cisco Packet Tracer
Logical
Physical
192.168.0.100
SMARTPHONE-PT
Smartphone0
192.168.0.109
Server-PT
Server0
192.168.0.103
Laptop-PT
Laptop0
HomeRouter-PT-AC
Wireless Router0
192.168.0.106
PC-PT
PC0
PC-PT
PC1
192.168.0.105
Time: 00:27:38
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
AccessPoint-PT-N

A) IPCONFIG

i) C:\>ipconfig

Wireless0 Connection:(default port)

Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::207:ECFF:FEC3:41B4
IPv6 Address.....: ::
IPv4 Address.....: 192.168.0.109
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
192.168.0.1

ii) C:\>ipconfig

Wireless0 Connection:(default port)

Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::201:42FF:FEA8:C03B
IPv6 Address.....: ::
IPv4 Address.....: 192.168.0.100
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
192.168.0.1

3G/4G Cell1 Connection:

Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::240:BFF:FE0E:EA80
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0

Bluetooth Connection:
--More--
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0

B) PING

i) C:\>ping 192.168.0.100

Pinging 192.168.0.100 with 32 bytes of data:

Reply from 192.168.0.100: bytes=32 time=25ms TTL=128
Reply from 192.168.0.100: bytes=32 time=17ms TTL=128
Reply from 192.168.0.100: bytes=32 time=17ms TTL=128
Reply from 192.168.0.100: bytes=32 time=15ms TTL=128

Ping statistics for 192.168.0.100:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 15ms, Maximum = 25ms, Average = 18ms

ii) C:\>ping 192.168.0.105

Pinging 192.168.0.105 with 32 bytes of data:

Reply from 192.168.0.105: bytes=32 time=30ms TTL=128
Reply from 192.168.0.105: bytes=32 time=16ms TTL=128
Reply from 192.168.0.105: bytes=32 time=16ms TTL=128
Reply from 192.168.0.105: bytes=32 time=18ms TTL=128

Ping statistics for 192.168.0.105:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 16ms, Maximum = 30ms, Average = 20ms